

École Centrale de Lille & Université de Lille

IRCICA: Internship Report

Hybrid-Precision Spatio-Temporal Attention

Design and Experimental Evaluation of HP-STAtten

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Abstract

Spiking Transformers promise the representational power of self-attention with the event-driven efficiency of Spiking Neural Networks (SNNs), yet existing designs force a difficult trade-off among temporal modeling, representational capacity, and computational cost. Spatial-Temporal Attention (STAtten) captures multi-step dynamics via temporal chunking and a low-rank factorization $\mathbf{A} = \mathbf{K}^\top \mathbf{V}$, but relies on binary spike encodings throughout. Addition-Only Spiking Self-Attention (A²OS²A) is reported to improve expressivity with hybrid-precision query-key-value (Q/K/V) encodings, yet its naive formulation scales as $\mathcal{O}(N^2)$ in the number of patches N . Spike-Driven Self-Attention (SDSA) reduces complexity to $\mathcal{O}(N \cdot D_h)$ via a Hadamard product, but does not model cross-channel interactions and lacks explicit spatio-temporal chunking.

This internship proposes **HP-STAtten** (*Hybrid-Precision STAtten*), which combines STAtten’s chunk-wise factorized attention with A²OS²A’s hybrid Q/K/V encodings, yielding $\mathcal{O}(N \cdot D_h^2)$ complexity per chunk while preserving addition-only, Softmax-free computation. We implement four baseline Spiking Transformers and HP-STAtten in a modular codebase, introduce a 4×2 ablation grid over attention structure (**factorized**, Hadamard/**sdt**, contrast, contrast_**sdt**) and Q/K/V precision (hybrid *vs.* binary), and evaluate on **CIFAR-10** and **CIFAR-10-DVS**. After Optuna tuning of the reference corner and transferring those hyperparameters at $D=512$ for 310 epochs across the full 4×2 CIFAR-10 grid, factorized, contrast, and contrast_**sdt** variants reach **~95.4 to 96.1%** accuracy on the official test set, while pure Hadamard (**sdt**) variants remain ~2.5 to 3.1 points lower (**~93.0 to 93.6%**). Under this protocol, binary Q/K/V matches or slightly exceeds hybrid encodings in every structure row, contrary to the hybrid advantage reported by A²OS²A in its $\mathcal{O}(N^2)$ setting, so attention *structure* remains the dominant accuracy axis. On CIFAR-10-DVS (partial grid: hybrid structure sweep plus binary factorized reference), hybrid contrast reaches **81.55%** (above STAtten+SDT, 2-256, ~81.1%), and hybrid **sdt** (81.15%) outperforms hybrid factorized (79.96%); binary factorized is 80.26%. Event-based structure rankings need not match static CIFAR-10.

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1 Introduction

1.1 Internship context and objectives

This report documents an internship conducted at IRCICA (Research Institute on Hardware-Software Components for Advanced Information and Communication Applications), in collaboration with École Centrale de Lille and Université de Lille. The project focuses on *Spiking Transformers* for image classification: architectures that combine the representational power of self-attention with the event-driven efficiency of Spiking Neural Networks (SNNs).

The internship pursued three objectives:

1. **Literature review and reimplementation.** Understand four recent Spiking Transformer architectures (Spikformer, Spike-Driven Transformer, STAtten, A²OS²A) through paper study and modular reimplementation.
2. **Methodological contribution.** Design and implement **HP-STAtten**, a hybrid-precision spatio-temporal attention module that combines ideas from STAtten and A²OS²A within a spike-driven backbone, extended with Hadamard and contrastive attention modes.
3. **Experimental evaluation.** Benchmark a 4×2 ablation grid on **CIFAR-10** and validate temporal modeling on **CIFAR-10-DVS**, under shared Optuna-informed recipes with tracked experiments.

This report uses both datasets: CIFAR-10 for the structure×precision ablation, and CIFAR-10-DVS to probe event-driven dynamics (temporal chunking applies to the factorized path; other modes remain per-timestep).

1.2 Background and motivation

Spiking Neural Networks (SNNs) communicate through sparse, discrete events over time, making them a natural fit for event-driven and neuromorphic hardware where energy is dominated by multiply-accumulate (MAC) operations rather than spike-triggered accumulations (AC) Roy et al. [2019]. As Transformers became the dominant paradigm in computer vision, a parallel line of work asked whether self-attention could be adapted to spike-based computation without sacrificing the accuracy of Vision Transformers (ViTs). *Spiking Transformers* Zhou et al. [2022], Yao et al. [2023] pursue this goal by replacing softmax attention with spike-compatible, addition-dominated alternatives while retaining the patch-based pipeline: Spiking Patch Splitting (SPS), stacked attention-MLP blocks, and a classification head.

Despite rapid progress, current Spiking Transformers still disagree on *how* to compute attention efficiently. Early Spikformer-style models Zhou et al. [2022] apply spiking self-attention (SSA) over flattened tokens with quadratic cost in the number of patches N . Spike-Driven Transformer (SDT) Yao et al. [2023] reformulates attention as a linear, Hadamard-based Spike-Driven Self-Attention (SDSA): $\mathbf{kv} = \sum(\mathbf{K} \odot \mathbf{V})$, $\mathbf{O} = \mathbf{Q} \odot \mathbf{kv}$, reducing spatial complexity to $\mathcal{O}(N \cdot D_h)$ but restricting cross-channel mixing to a diagonal interaction. STAtten Wang et al. [2024] introduces explicit *temporal chunking* and a

factorized map $\mathbf{A} = \mathbf{K}^\top \mathbf{V} \in \mathbb{R}^{D_h \times D_h}$ per head, yielding $\mathcal{O}(N \cdot D_h^2)$ cost that is sub-quadratic in N while enabling richer inter-channel context within each chunk. More recently, A²OS²A Li et al. [2025] shows that Q, K, and V need not share the same precision: binary spikes for queries, floating-point keys, and ternary values can improve representational capacity under an addition-only, softmax-free attention, but at $\mathcal{O}(N^2)$ in its naive form. Orthogonal work on Visual-Contrast Attention Pu et al. [2025] suggests that differential interactions over pooled contrast tokens can be more discriminative than similarity alone, at complexity comparable to factorized maps when $n \ll N$.

These advances are complementary yet fragmented. STAtten models time and structure efficiently but keeps Q/K/V binary; A²OS²A enriches encoding but ignores STAtten’s temporal factorization; SDT minimizes FLOPs but neither chunks time nor supports hybrid-precision encodings; contrastive linear attention has not been studied inside a controlled Spiking Transformer grid. No existing method provides a *unified, ablatable* framework that jointly controls (i) attention structure, (ii) Q/K/V precision, and (iii) computational complexity. Moreover, complexity (FLOPs/MACs) and energy (AC operations, spike sparsity) are often conflated in the literature.

1.3 Contributions

We address this gap with **HP-STAtten** (*Hybrid-Precision Spatial-Temporal Attention*), a Spiking Transformer attention module that combines STAtten’s chunk-wise factorization with A²OS²A’s hybrid Q/K/V encoding inside the spike-driven backbone shared with SDT. Concretely, within each temporal chunk we compute

$$\mathbf{A}_\tau = \mathbf{K}_\tau^\top \mathbf{V}_\tau, \quad \mathbf{O}_\tau = \text{SN}(\mathbf{Q}_\tau \mathbf{A}_\tau), \quad (1)$$

where $\mathbf{Q}_\tau \in \{0, 1\}^{N \times D_h}$, $\mathbf{K}_\tau \in \mathbb{R}_+^{N \times D_h}$, and $\mathbf{V}_\tau \in \{-1, 0, 1\}^{N \times D_h}$, i.e. Softmax-free *right-to-left* association ($\mathbf{K}^\top \mathbf{V}$ first, then $\mathbf{Q}$), in contrast to Softmax MHSA / original VCA which form token affinities left-to-right. We additionally expose a 4×2 ablation grid over attention structure ($\{\text{factorized, Hadamard/sdt, contrast, contrast_sdt}\}$) and Q/K/V precision ($\{\text{hybrid, binary}\}$).

Our contributions are threefold:

1. **HP-STAtten.** A spatio-temporal, hybrid-precision attention operator with $\mathcal{O}(N \cdot D_h^2)$ complexity per chunk, combining STAtten’s factorization with A²OS²A’s hybrid Q/K/V encodings without the $\mathcal{O}(N^2)$ cost of naive hybrid attention.
2. **Ablatable 4×2 design space with Softmax-free integrations.** A single implementation that isolates attention *structure* (factorized / Hadamard / contrast / contrast_sdt) and *precision* (hybrid *vs.* binary), including Softmax-free right-to-left redesigns of VCA’s originally left-to-right Softmax stages, under a shared training recipe.
3. **CIFAR-10 and CIFAR-10-DVS evidence.** With Optuna trial #27 hyper-parameters transferred across the full 4×2 CIFAR-10 grid ($D = 512$, 310 epochs, seed 42), factorized, contrast, and **contrast_sdt** variants reach ~ 95.4 to 96.1% test accuracy, while pure Hadamard (**sdt**) variants lag by ~ 2.5 to 3.1 points (~ 93.0 to

93.6%). Contrary to A²OS²A’s reported hybrid advantage, binary Q/K/V matches or slightly exceeds hybrid in every structure row under this protocol. On CIFAR-10-DVS (partial grid), the STAtten-like corner reaches 80.26%; a hybrid structure sweep reaches **81.55%** with contrast (above STAtten+SDT, 2-256, ~81.1%), while hybrid **sdt** (81.15%) outperforms hybrid factorized (79.96%), so the CIFAR-10 structure ranking does not transfer to event streams.

1.4 Report organization

Section 2 reviews related work on Spiking Transformers and positions HP-STAtten in the design space. Section 3 describes the HP-STAtten module, the 4×2 ablation grid, complexity analysis, and the implementation framework. Section 4 presents datasets, the CIFAR-10 ablation, CIFAR-10-DVS validation, and complexity-accuracy interpretation. Section 5 summarizes limitations and perspectives.

2 Related Work

We review Spiking Transformers through the lens of *attention formulation*, *temporal modeling*, and *encoding precision*, the three axes that HP-STAtten unifies.

2.1 Spiking Transformers and linear attention

Spikformer Zhou et al. [2022] was among the first to combine Vision Transformers with SNNs, replacing softmax attention with Spiking Self-Attention (SSA) over token sequences. SSA preserves a quadratic $\mathcal{O}(N^2)$ interaction in the number of spatial patches N , but operates on binary spike trains and avoids explicit softmax normalization.

Spike-Driven Transformer (SDT) Yao et al. [2023] takes a different route: attention is computed directly on 2D spike feature maps (T, B, D, H, W) using Spike-Driven Self-Attention (SDSA),

$$\mathbf{kv} = \sum_{n=1}^N \mathbf{K}_n \odot \mathbf{V}_n, \quad \mathbf{O} = \mathbf{Q} \odot \mathbf{kv}, \quad (2)$$

where $\odot$ denotes the Hadamard product and $\mathbf{kv} \in \mathbb{R}^{D_h}$ is a per-head context vector broadcast over tokens. SDSA reduces spatial complexity to $\mathcal{O}(N \cdot D_h)$ and, when Q/K/V are binary, enables mask-and-add implementations favorable to neuromorphic hardware. Its limitation is structural: each output channel depends only on the matching channel in $\mathbf{V}$; there is no cross-channel mixing within the attention map. SDT also shares with Spikformer a largely *spatial* treatment of time: temporal steps T are stacked but not explicitly chunked inside attention.

2.2 Spatio-temporal attention

STAtten Wang et al. [2024] addresses temporal structure by grouping T into non-overlapping chunks of size c and performing attention on the $c \cdot N$ tokens of each chunk. Instead of materializing an $(cN \times cN)$ affinity matrix, STAtten factorizes attention per head as

$$\mathbf{A} = \mathbf{K}^\top \mathbf{V} \in \mathbb{R}^{D_h \times D_h}, \quad \mathbf{O} = \mathbf{Q} \mathbf{A}, \quad (3)$$

yielding $\mathcal{O}(c \cdot N \cdot D_h^2)$ cost per chunk, which is sub-quadratic in N for typical head dimensions $D_h \ll N$. This factorization is strictly more expressive than SDSA: $\mathbf{A}$ mixes all value channels before projecting onto queries. STAtten keeps Q/K/V as binary spikes throughout, trading representational flexibility for spike sparsity.

2.3 Hybrid-precision spiking attention

A²OS²A Li et al. [2025] argues that uniform binarization of Q/K/V is unnecessary. Queries are encoded as binary LIF spikes $\mathbf{Q} \in \{0, 1\}$, keys as ReLU floats $\mathbf{K} \in \mathbb{R}_+$, and values as ternary LIF outputs $\mathbf{V} \in \{-1, 0, 1\}$. Attention becomes addition-only: $\mathbf{O} = \text{SN}(\mathbf{Q} \mathbf{K}^\top \mathbf{V})$ without softmax or scaling. This hybrid design is reported to increase expressivity over all-binary encodings in their $\mathcal{O}(N^2)$ setting, but the naive token form is impractical at image resolutions where N is large. Whether the same hybrid advantage

transfers to STAtten-style right-to-left factorization ($\mathbf{O} = \mathbf{Q}(\mathbf{K}^\top \mathbf{V})$) is precisely one axis of our 4×2 ablation (Section 4).

2.4 Visual-contrast attention

Outside SNNs, Visual-Contrast Attention (VCA) Pu et al. [2025] replaces Softmax MHSA with a differential mechanism over pooled contrast tokens ($n \ll N$), arguing that positive/negative differences carry stronger discriminative signal than raw similarity. Importantly, original VCA remains *left-to-right* Softmax attention ($\text{Softmax}(\mathbf{t}\mathbf{K}^\top)\mathbf{V}$, then $\text{Softmax}(\mathbf{Q}\mathbf{t}^\top)\hat{\mathbf{v}}$). Our Softmax-free VCA-light paths (`contrast`, `contrast_sdt`) keep the contrastive bias but rewrite both stages in the STAtten/SDT *right-to-left* associative form (Sections 3.5 and 3.7).

2.5 Positioning of HP-STAtten

Table 1 summarizes how prior methods occupy the design space HP-STAtten spans. No prior work jointly exposes STAtten-style chunked factorization, A²OS²A-style hybrid encodings, and an SDT-style linear Hadamard path inside one Spiking Transformer module; our 4×2 grid (Section 3.8) further adds Softmax-free contrastive corners for controlled comparison.

Table 1: Positioning of prior methods and ours (N patches, head dimension D_h). **Chunk**: temporal chunking; **Hybrid**: hybrid Q/K/V encoding; **Cost**: asymptotic complexity. [†]Hadamard (linear) variant without chunk-wise factorization.

Method	Chunk	Hybrid	Cost
Spikformer Zhou et al. [2022]	✗	✗	$\mathcal{O}(N^2)$
SDT Yao et al. [2023]	✗	✗	$\mathcal{O}(ND_h)$
STAtten Wang et al. [2024]	✓	✗	$\mathcal{O}(ND_h^2)$
A ² OS ² A Li et al. [2025]	✗	✓	$\mathcal{O}(N^2)$
HP-STAtten	✓	✓	$\mathcal{O}(ND_h^2)$
HP-STAtten-lin. [†]	✗	✓	$\mathcal{O}(ND_h)$

2.6 Complexity versus energy

Following Yao et al. [2023], Li et al. [2025], we distinguish *computational complexity* (FLOPs/MACs of the attention mixing step) from *hardware energy* (MAC *vs.* AC operations under spike sparsity). A method can be FLOP-efficient yet still MAC-heavy if activations are dense floats; conversely, binary Q/K/V can shift multiplications to cheaper accumulations even when FLOP counts are similar. Importantly, STAtten-style factorization already eliminates the Softmax N^2 term: the remaining structural bottleneck is D_h^2 channel mixing, not patch-patch affinity. Our 4×2 grid separates these axes: rows

control structural FLOPs and inductive bias; columns control the MAC/AC mix via hybrid *vs.* binary encodings (Section 3.9).

3 Method

We present HP-STAtten as a drop-in attention module inside the spike-driven Transformer backbone shared with SDT Yao et al. [2023]. Notation: input spike maps $\mathbf{X} \in \{0, 1\}^{T \times B \times D \times H \times W}$; $N = H \cdot W$ patches; H_h heads with $D_h = D/H_h$; n contrast tokens when using `contrast` or `contrast_sdt` mode.

3.1 Backbone overview

The network follows the standard Spiking Transformer pipeline: **(1)** Spiking Patch Splitting (SPS) converts the input into multi-step feature maps; **(2)** a stack of L blocks, each consisting of attention + MLP with LIF activations and residual connections; **(3)** a classification head that applies global average pooling over space, a final LIF, a linear layer, and temporal averaging over T . We adopt SDT’s convolutional MLP (LIF $\rightarrow$ Conv $\rightarrow$ LIF) and 2D feature-map format, which avoids flattening patches into long token sequences and matches STAtten’s spatial layout. Figure 1 gives an overview of the full pipeline; Figures 2 and 4 detail the patch embedding and feed-forward blocks.

OVERALL ARCHITECTURE

HP-STAtten Spiking Transformer

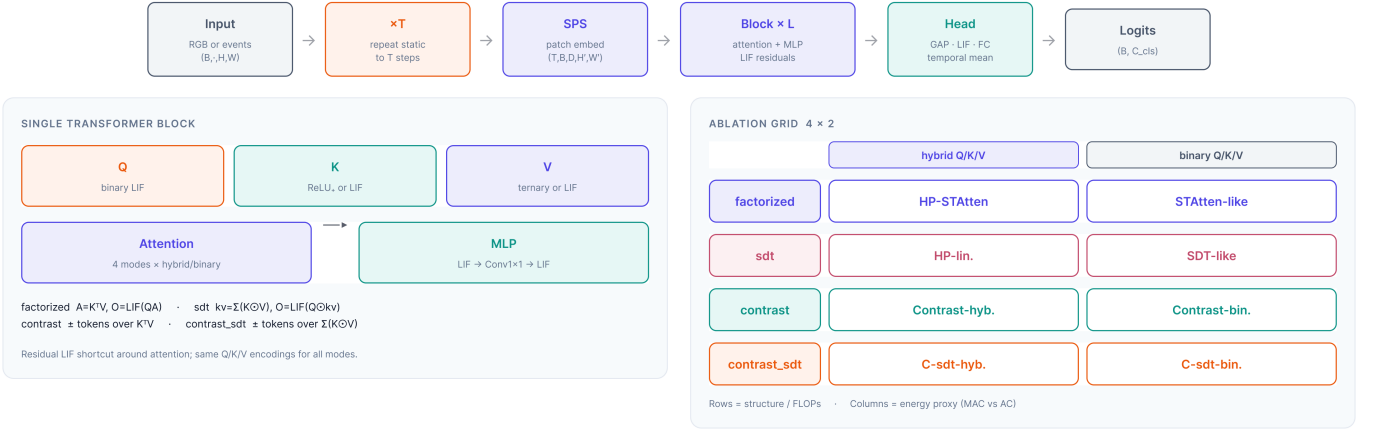


Figure 1: Overall architecture of the HP-STAtten Spiking Transformer. Static RGB inputs are repeated over T timesteps (event streams already have a temporal axis). Spiking Patch Splitting (SPS) yields feature maps (T, B, D, H', W') ; L residual blocks apply attention (hybrid or binary Q/K/V per ablation) and a spike-driven MLP; the head pools spatially then temporally. Right: the 4×2 ablation over attention structure and Q/K/V precision.

3.2 Spiking patch embedding

The Spiking Patch Splitting (SPS) module maps raw inputs to spike feature maps of shape (T, B, D, H', W') . As shown in Figure 2, four stacked conv-BN-LIF-MaxPool

stages increase the channel width while reducing spatial resolution; a relative position encoding (RPE) branch is fused residually at the output.

PATCH EMBEDDING

Spiking Patch Splitting (SPS)

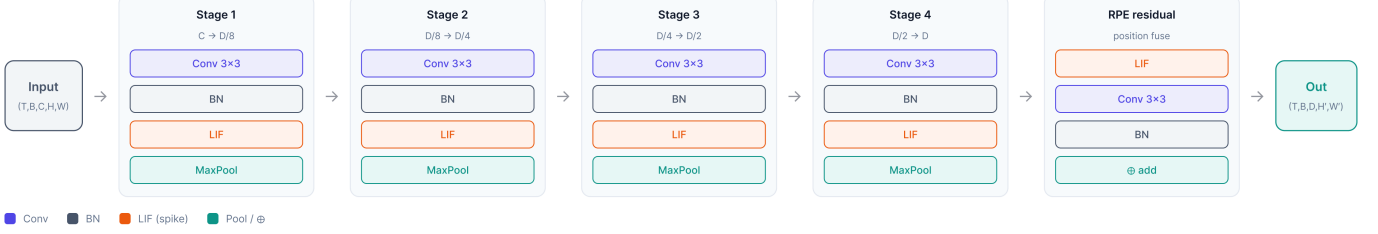


Figure 2: Spiking Patch Splitting (SPS). Four Conv-BN-LIF-MaxPool stages expand channels $C \rightarrow D/8 \rightarrow D/4 \rightarrow D/2 \rightarrow D$ while downsampling space. A relative position encoding (RPE) branch is fused residually before the block stack.

3.3 Hybrid-precision Q/K/V encoding

Given feature maps $\mathbf{X}_\tau$ at time τ , linear projections produce pre-activations that are encoded differently depending on the `hybrid_qkv` flag:

$$\mathbf{Q}_\tau = \text{LIF}(\text{Conv}(\mathbf{X}_\tau)) \in \{0, 1\}^{N \times D_h}, \quad (4)$$

$$\mathbf{K}_\tau = \text{ReLU}(\text{Conv}(\mathbf{X}_\tau)) \in \mathbb{R}_+^{N \times D_h}, \quad (5)$$

$$\mathbf{V}_\tau = \text{TernaryLIF}(\text{Conv}(\mathbf{X}_\tau)) \in \{-1, 0, 1\}^{N \times D_h}, \quad (6)$$

when `hybrid_qkv=true` (A²OS²A-style). Setting `hybrid_qkv=false` replaces ReLU and TernaryLIF with standard LIF spikes, recovering binary Q/K/V as in STAtten Wang et al. [2024]. All encodings are computed per head; a shortcut LIF branch and output LIF SN($\cdot$) follow A²OS²A’s addition-only design, without softmax or temperature scaling. Figure 3 illustrates the attention module (factorized, Hadamard, and contrast paths share the same hybrid/binary Q/K/V encodings).

3.4 Factorized spatio-temporal attention

Time is partitioned into T/c non-overlapping chunks of size $c = \text{chunk_size}$. Within chunk τ , tokens from all c timesteps are concatenated ($c \cdot N$ tokens total) and attention is computed as in Equation (1):

$$\mathbf{A}_\tau = \mathbf{K}_\tau^\top \mathbf{V}_\tau \in \mathbb{R}^{D_h \times D_h}, \quad \mathbf{O}_\tau = \text{SN}(\mathbf{Q}_\tau \mathbf{A}_\tau). \quad (7)$$

This is the default `attention_mode=factorized` path and matches STAtten’s factorization while preserving hybrid encodings. Per chunk and head, complexity is $\mathcal{O}(c \cdot N \cdot D_h^2)$, linear in N and quadratic only in D_h ($D_h \in [32, 64]$ in our setups).

ATTENTION MODULE

Hybrid-precision encodings and four mixing modes



Color encodes role: orange = spikes / LIF · indigo = factorized mix · teal = contrast / residual · rose = Hadamard. Softmax is never used.

Figure 3: HP-STAtten attention module (Softmax-free). **(a)** Hybrid or binary Q/K/V encodings shared by all modes. **(b)** Factorized STAtten map $A = K^T V$, complexity $\mathcal{O}(ND_h^2)$. **(c)** SDT Hadamard (diagonal of (b)), $\mathcal{O}(ND_h)$. **(d)/(e)** VCA-light contrast with factorized vs. linear aggregation. **(f)** Projection and residual with a LIF shortcut.

3.5 Association order: right-to-left versus Softmax left-to-right

A subtle but central design choice separates our integrations from Softmax Vision Transformers (and from original VCA).

Left-to-right (token affinity first). Classical MHSA evaluates

$$\mathbf{O} = \text{Softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{D_h}) \mathbf{V}, \quad (8)$$

i.e. an $N \times N$ token-token map *before* mixing values. VCA Pu et al. [2025] keeps the same *left-to-right* order with contrast mediators: $\text{Softmax}(\mathbf{t}\mathbf{K}^\top)\mathbf{V}$ then $\text{Softmax}(\mathbf{Q}\mathbf{t}^\top)\hat{\mathbf{v}}$. In both cases the dominant object is a (possibly reduced) spatial affinity matrix, followed by Softmax densification.

Right-to-left (channel map first). STAtten and HP-STAtten instead use associativity without Softmax:

$$\mathbf{O} = \mathbf{Q}(\mathbf{K}^\top \mathbf{V}), \quad (9)$$

forming a $D_h \times D_h$ *channel* map from keys and values, then querying it with $\mathbf{Q}$. Computation proceeds **right-to-left**: aggregate $\mathbf{K}, \mathbf{V}$ first, apply $\mathbf{Q}$ second. This removes the $N \times N$ Softmax while retaining cross-channel mixing, and is addition-friendly under spike/hybrid encodings.

Why the rewrite matters for our work. All four attention modes in HP-STAtten are built in this right-to-left regime: **factorized** uses $\mathbf{K}^\top \mathbf{V}$; **sdt** keeps only its diagonal (Hadamard); **contrast** / **contrast_sdt** transplant VCA’s differential $\pm$ bias onto the same associative backends (Section 3.7). We therefore do *not* reintroduce $\text{Softmax}(\mathbf{Q}\mathbf{K}^\top)$, even when borrowing ideas from left-to-right papers: the inductive bias is kept, the association order is changed.

3.6 Linear Hadamard variant

When `attention_mode=sdt`, we replace the factorized map with SDSA-style Hadamard attention (Equation (2)) on the same Q/K/V encodings:

$$\text{kv}_\tau = \sum_{n=1}^N \mathbf{K}_{\tau,n} \odot \mathbf{V}_{\tau,n}, \quad \mathbf{O}_\tau = \text{SN}(\mathbf{Q}_\tau \odot \text{SN}(\text{kv}_\tau)), \quad (10)$$

where $\text{kv}_\tau \in \mathbb{R}^{D_h}$ is broadcast over the N spatial tokens. An intermediate *talking-heads* LIF is applied to kv_τ before the Hadamard product with $\mathbf{Q}_\tau$, following SDT Yao et al. [2023]. Complexity drops to $\mathcal{O}(N \cdot D_h)$ per timestep; cross-channel mixing is lost because each channel is updated independently.

SDSA as a diagonal restriction. The factorized map $\mathbf{O} = \mathbf{Q}(\mathbf{K}^\top \mathbf{V})$ computes output channel i as $\mathbf{Q}_i \sum_j (\mathbf{K}^\top \mathbf{V})_{ij}$, mixing all value channels j . SDSA computes output channel i as $\mathbf{Q}_i \sum_n K_{i,n} V_{i,n}$, the diagonal of the same interaction. SDSA is therefore a lower-expressivity, lower-cost special case of the *same* right-to-left product, not an unrelated left-to-right mechanism.

3.7 Contrastive (VCA-light) variants

We adapt Visual-Contrast Attention Pu et al. [2025] to the SNN setting (*VCA-light*). The subtlety is that we keep VCA’s *contrastive inductive bias* while changing its *association order*.

What we keep from VCA. Queries are spatially pooled into $n \ll N$ contrast tokens $\mathbf{t}$, then split into positive/negative streams $\mathbf{t}_\pm = \mathbf{t} + \mathbf{e}_\pm$ with learned positional embeddings $\mathbf{e}_\pm$. A two-stage differential update ($\hat{\mathbf{v}}$ from $\mathbf{t}_\pm$, then $\mathbf{O}$ from $\mathbf{Q}$ and $\hat{\mathbf{v}}$) implements “difference over similarity”.

What we change. Original VCA is left-to-right and Softmax-based: $\text{Softmax}(\mathbf{t}\mathbf{K}^\top)\mathbf{V}$ then $\text{Softmax}(\mathbf{Q}\mathbf{t}^\top)\hat{\mathbf{v}}$. Because Softmax breaks associativity, VCA-light does *not* preserve that operator: it redesigns both stages as Softmax-free, LIF-gated, right-to-left maps in the STAtten/SDT style, and reuses the same hybrid/binary Q/K/V flags as the rest of the grid. Two aggregation backends share this token construction and differ only in how context is mixed.

contrast (factorized aggregation). With $\mathbf{KV} = \mathbf{K}^\top \mathbf{V}$, Stage I becomes $\mathbf{t}_\pm \mathbf{KV}$ (instead of $\text{Softmax}(\mathbf{t}_\pm \mathbf{K}^\top)\mathbf{V}$) and Stage II becomes $\mathbf{Q}(\mathbf{t}_\pm^\top \hat{\mathbf{v}})$:

$$\hat{\mathbf{v}} = \text{SN}(\mathbf{t}_+ \mathbf{KV} - \lambda_1 \mathbf{t}_- \mathbf{KV}), \quad \mathbf{O} = \text{SN}(\mathbf{Q}(\mathbf{t}_+^\top \hat{\mathbf{v}} - \lambda_2 \mathbf{t}_-^\top \hat{\mathbf{v}})). \quad (11)$$

Per-head complexity is $\mathcal{O}((N+n)D_h^2)$, the same asymptotic family as factorized attention, with a small constant overhead from the dual streams ($n=16$ vs. $N \approx 64$ on CIFAR-10 after SPS).

contrast_sdt (linear aggregation). Replacing $\mathbf{K}^\top \mathbf{V}$ with SDSA-style Hadamard aggregation yields the linear twin (again right-to-left, now diagonal):

$$\begin{aligned} \mathbf{kv} &= \text{SN}\left(\sum_n \mathbf{K}_n \odot \mathbf{V}_n\right), \\ \hat{\mathbf{v}} &= \text{SN}((\mathbf{t}_+ \odot \mathbf{kv} - \lambda_1 \mathbf{t}_- \odot \mathbf{kv})), \\ \mathbf{O} &= \mathbf{Q} \odot (\sum(\mathbf{t}_+ \odot \hat{\mathbf{v}}) - \lambda_2 \sum(\mathbf{t}_- \odot \hat{\mathbf{v}})). \end{aligned} \quad (12)$$

Complexity falls to the $\mathcal{O}((N+n)D_h)$ family, mirroring **sdt** while retaining the contrastive inductive bias. This closes the design space: contrastive attention exists in both the factorized and linear *right-to-left* regimes.

3.8 Ablatable 4×2 design grid

$\text{attention_mode} \in \{\text{factorized}, \text{sdt}, \text{contrast}, \text{contrast_sdt}\}$ and $\text{hybrid_qkv} \in \{\text{true}, \text{false}\}$ yield eight interpretable corners under identical code and training (Table 2).

Table 2: 4×2 ablation grid ($\text{attention_mode} \times \text{hybrid_qkv}$). Rows: attention structure; columns: Q/K/V precision. Asymptotic attention cost: $\text{sdt}/\text{contrast_sdt}$ $\mathcal{O}(ND_h)$ family; $\text{factorized}/\text{contrast}$ $\mathcal{O}(ND_h^2)$ family.

	Hybrid Q/K/V	Binary Q/K/V
factorized	HP-STAtten	STAtten-like
sdt	HP-STAtten-lin.	SDT-like
contrast	Contrast-hyb.	Contrast-bin.
contrast_sdt	Contrast-sdt-hyb.	Contrast-sdt-bin.

This grid isolates two independent levers: **(i) structure**: full $D_h \times D_h$ mixing (factorized), diagonal Hadamard (**sdt**), differential contrast with factorized aggregation, or differential contrast with linear aggregation; **(ii) precision**: hybrid Q/K/V *vs.* all-binary spikes. **hybrid_qkv** does *not* change the FLOP order of the mixing step; it mainly changes the MAC/AC mix (energy proxy on neuromorphic hardware). All eight corners share the same SPS, MLP, head, and training recipe (Section 4). Temporal chunking (concatenating c timesteps into cN tokens before mixing) is used only for **attention_mode=factorized**; **sdt**, **contrast**, and **contrast_sdt** mix per timestep.

3.9 Complexity analysis

We analyze **attention mixing** only (one head, one timestep or chunk). Q/K/V projections and SPS are shared across corners and therefore cancel in relative comparisons. Notation matches our primary CIFAR-10 ablation after SPS with **pooling_stat=0011**: $N \approx 64$ spatial tokens, $C=512$, $M=8$ heads, $D_h=64$, $n=16$ contrast tokens, $T=4$ (DVS baselines use $C=256$, $D_h=32$, $T=16$).

Structural cost by attention_mode. Table 3 summarizes the four structures. Classical Softmax MHSA scales as $\mathcal{O}(N^2 D_h)$ per head; STAtten-style factorization already removes the N^2 term via right-to-left associativity ($\mathbf{A} = \mathbf{K}^\top \mathbf{V} \in \mathbb{R}^{D_h \times D_h}$; Section 3.5), leaving $\mathcal{O}(ND_h^2)$, quadratic only in the small head dimension. **sdt** further restricts $\mathbf{A}$ to its diagonal (Hadamard), dropping to $\mathcal{O}(ND_h)$. **contrast** stays in the factorized family with a dual $\pm$ stream over n tokens; **contrast_sdt** is the corresponding linear twin. Consequently, transplanting VCA into this regime does *not* inherit VCA’s complexity win over Softmax MHSA: that win assumes a left-to-right Softmax baseline we have already left.

At our CIFAR resolution, Softmax MHSA and **factorized** happen to sit near the same numeric order ($\sim 10^5$); as N grows (e.g., weaker pooling or higher-resolution DVS),

Table 3: Attention-mixing complexity (one head). Layer cost multiplies by M heads ($\equiv$ replace D_h by C in leading terms). Order-of-magnitude estimates use $N \approx 64$, $D_h = 64$, $n = 16$ (CIFAR-10 $D = 512$).

Mode	Per head	Layer (M heads)	Approx. MACs (1 head)
sdt	$\mathcal{O}(ND_h)$	$\mathcal{O}(NC)$	$\sim 4 \times 10^3$
contrast_sdt	$\mathcal{O}((N+n)D_h)$	$\mathcal{O}((N+n)C)$	$\sim 5 \times 10^3$
factorized	$\mathcal{O}(ND_h^2)$	$\mathcal{O}(NCD_h)$	$\sim 2.6 \times 10^5$
contrast	$\mathcal{O}((N+n)D_h^2)$	$\mathcal{O}((N+n)CD_h)$	$\sim 3.3 \times 10^5$
Softmax MHSA	$\mathcal{O}(N^2D_h)$	$\mathcal{O}(N^2C)$	$\sim 2.6 \times 10^5$

MHSA scales as N^2 while our modes remain linear in N . Within the grid,

$$\text{sdt} \lesssim \text{contrast_sdt} \ll \text{factorized} \approx \text{contrast} \ll \text{MHSA } (N \uparrow).$$

Precision vs. energy (orthogonal axis). `hybrid_qkv` does *not* change the asymptotic FLOP order of mixing. It changes the tensor types (binary LIF queries in both cases; ReLU float keys and ternary values when hybrid; all-binary LIF when not) and therefore the expected MAC/AC mix on neuromorphic hardware. Binary corners are the lower energy proxy; hybrid corners trade some AC-friendliness for denser keys. Training-GPU wall-clock is typically dominated by SPS and dense kernels, not by this proxy, so we treat energy as a *theoretical* axis separated from FLOP complexity (Section 2.6).

What the grid isolates. Comparing within a column holds precision fixed and varies structure (inductive bias and FLOPs). Comparing within a row holds structure fixed and varies hybrid *vs.* binary (energy proxy). Comparing **contrast** to **factorized** (same column) tests whether a differential bias helps at comparable $\mathcal{O}(ND_h^2)$ cost; comparing **contrast_sdt** to **sdt** asks the same question in the linear regime.

Orthogonal complexity ablations (out of scope for claims). Separately from the 4×2 grid, the codebase exposes optional factorized mixers (local window, low-rank rank- r , Nyström landmarks) that attack residual D_h^2 cost or spatial bias on the HP-STAtten reference corner only. These are exploratory and not used in the accuracy claims of Section 4.

3.10 Spike-driven MLP block

Each transformer block includes a convolutional MLP following the SDT design (Figure 4): two sequential stages expand then contract channels ($C \rightarrow C_h \rightarrow C$), each gated by a binary LIF then Conv1 $\times$ 1 and BatchNorm, with residual shortcuts.

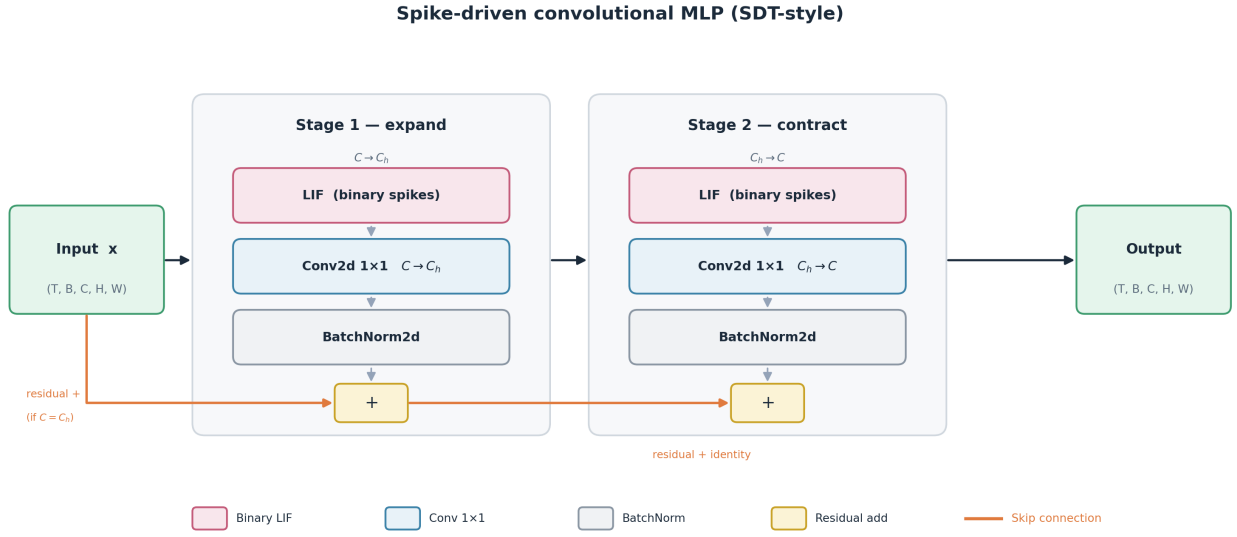


Figure 4: Spike-driven convolutional MLP block (SDT-style). Two sequential stages expand then contract channels ($C \rightarrow C_h \rightarrow C$): each applies binary LIF $\rightarrow$ Conv $1 \times 1 \rightarrow$ BatchNorm, with residual skip connections (orange).

3.11 Implementation notes

HP-STAtten is implemented in PyTorch with SpikingJelly LIF neurons. Temporal talking-heads LIF is used only in `sdt` mode; contrast modes share a common mixer, with `contrast_sdt` enabling linear aggregation. Experiments are tracked with MLflow on CIFAR-10 and CIFAR-10-DVS under a shared training stack (augmentation, schedulers, and logging).

4 Experiments and Results

4.1 Experimental protocol

4.1.1 Datasets

We evaluate on two complementary benchmarks that play distinct roles in the study.

CIFAR-10 (static RGB). CIFAR-10 Krizhevsky [2009] provides 50,000 training and 10,000 test 32×32 RGB images over 10 classes. We encode each image as a $T=4$ -step spike stream (direct encoding through the SPS stack) and report **best accuracy on the official CIFAR-10 test set** (logged as `val_acc` in MLflow; no separate validation split). This dataset is our primary setting for the structure $\times$ precision ablation: it is fast enough for Optuna and long grid runs, while remaining standard in the Spiking Transformer literature.

CIFAR-10-DVS (event-based). CIFAR-10-DVS Li et al. [2017] converts the same 10-class taxonomy into neuromorphic event streams recorded with a DVS camera. Following SpikingJelly practice and the STAtten/SDT evaluation protocol, we use pre-integrated frames with $T = 16$ timesteps (`split_by=number`), resize to 64×64 , and a stratified **90 %/10 %** train/hold-out split (9,000/1,000). We report best accuracy on the 10 % hold-out (the metric used by SDT and STAtten), so that numbers are comparable to published 2-256, $T=16$ baselines. CIFAR-10-DVS probes *event-driven* dynamics that static CIFAR-10 cannot; temporal chunking applies to the factorized path, while other modes remain per-timestep.

Metrics and logging. All runs are logged with MLflow (train/val loss and accuracy per epoch; best checkpoint). Unless noted, we cite **best official-test (CIFAR-10) or hold-out (CIFAR-10-DVS) accuracy** over the full schedule.

4.1.2 Architecture

Unless noted, CIFAR-10 ablation models use embed dimension $D=512$, depth $L=2$, $H_h=8$ heads ($D_h=64$), chunk size $c=2$, and $n=16$ contrast tokens when applicable, matching the official STAtten CIFAR 2-512 width. CIFAR-10-DVS baselines in this section use $D=256$ (2-256) for comparability with published SDT/STAtten figures. The classification head follows SDT: spatial GAP, LIF, linear layer, temporal mean (TET-style averaging on DVS when configured).

4.1.3 CIFAR-10 training methodology

Phase 1 (baseline). Early runs (~ 15 epochs, no augmentation) landed near $\sim 80\%$ for all variants, far below published Spiking Transformer reports (~ 94 to 96%), confirming that training recipe (not only attention design) dominates short CIFAR-10 runs.

Phase 2 (literature-aligned recipe). We aligned training with recent Spiking Transformer practice: RandAugment, random erasing, label smoothing, cosine schedule, AdamW, and a long schedule.

Phase 3 (Optuna on the reference, then scaled transfer). We ran Optuna on HP-STatten (factorized, hybrid_qkv=true) and retained trial #27: $lr \approx 1.30 \times 10^{-3}$, weight decay $\approx 1.44 \times 10^{-4}$, mixup 0.2, label smoothing 0.15, cosine schedule, seed 42, full training set. Those optimizer/regularization hyperparameters were **frozen** and applied to all eight corners of the 4×2 grid in a scaled transfer campaign: $D=512$, 310 epochs, batch size 64, warmup 20 epochs (cifar10_grid_sota_512).

Methodological caveat. This protocol measures architecture under a shared recipe, not each corner at its own Optuna optimum. LR, mixup, and weight decay can interact with architecture; the residual **sdt** gap may therefore mix structural limits with HP mismatch. Per-corner Optuna (Option A) remains useful to isolate HP mismatch on linear corners.

Figure 5 summarizes the tooling stack used throughout the internship: datasets and DataLoader, Optuna for hyperparameter search, training/validation with checkpoints, and MLflow logging for analysis in the report and slides.

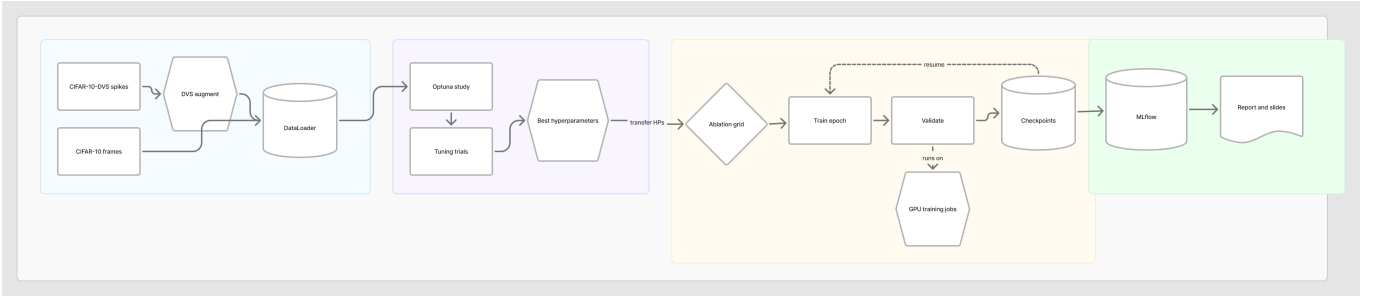


Figure 5: Project workflow and tooling. Datasets feed a DataLoader; Optuna searches hyperparameters; selected settings drive training and validation of model variants with checkpointing; MLflow logs metrics used for the report and slides.

MLflow run names for the CIFAR-10 4×2 campaign ($D=512$, 310 epochs): hpstattn-20260813-003627 (statten), hpstattn-20260811-171204 (hp), hpstattn-20260812-013354 (contrast), hpstattn-20260813-085723 (contrast_binary), hpstattn-20260811-212340 (hp_linear / sdt hybrid), hpstattn-20260813-044240 (sdt), hpstattn-20260812-202608 (contrast_sdt hybrid), hpstattn-20260813-085910 (contrast_sdt binary).

4.2 CIFAR-10 ablation: structure $\times$ precision

Table 4 reports best official-test accuracy for all eight corners of the 4×2 grid under the $D=512$, 310-epoch transfer. Two tiers appear:

- **Tier 1** (factorized / contrast / contrast_sdt): **95.36 to 96.10%**;

- **Tier 2** (Hadamard / **sdt**): **92.96 to 93.56%** (~ 2.5 to 3.1 points below the Tier 1 peak).

Within each structure row, the **binary** corner matches or exceeds its hybrid twin (gaps $+0.01$ to $+0.60$ points; mean $\sim +0.3$). Under transferred HPs at this scale, **structure dominates precision**, and the A^2OS^2A -style hybrid encoding does *not* improve accuracy over all-binary Q/K/V, despite Optuna having been tuned on the hybrid reference. The previous $D=256/200$ -epoch third tier for **contrast_sdt** disappears: the linear contrast twin joins the high-accuracy family once width and schedule are increased.

Table 4: CIFAR-10 4×2 ablation (Optuna t27 HPs transferred, $D = 512$, 310 epochs, seed 42). Best official-test accuracy from local metrics logs / MLflow (**val_acc**).

Rank	Variant	Mode	Hybrid	Best test (%)
1	STAtten-like	factorized	✗	96.10
2	Contrast-bin.	contrast	✗	95.65
3	Contrast-hyb.	contrast	✓	95.64
4	HP-STAtten	factorized	✓	95.57
5	Contrast-sdt-bin.	contrast_sdt	✗	95.47
6	Contrast-sdt-hyb.	contrast_sdt	✓	95.36
7	SDT-like	sdt	✗	93.56
8	HP-STAtten-lin.	sdt	✓	92.96

Comparison to published Spiking Transformers. Table 5 situates our best corners against the reference papers cited below (Spikformer Zhou et al. [2022], SDT Yao et al. [2023], STAtten Wang et al. [2024], A^2OS^2A Li et al. [2025]). All cited numbers are from-scratch CIFAR-10 results with $T=4$ unless noted. Our STAtten-like corner (96.10%) matches STAtten+SDT (96.03%) and sits ~ 0.3 points below A^2OS^2A ’s 2-512 hybrid model (96.42%), while contrast and HP-STAtten remain in the mid-95% band of Spikformer/SDT-scale models. Caveat: architectures, epochs (ours 310 vs. typically 200), and augmentation stacks are not identical: the table is a literature anchor, not a perfectly controlled bake-off.

[†]Longer training (400 epochs) reported by Spikformer.

4.3 Learning curves

Figure 6 shows official-test accuracy, train loss, and validation loss for all eight CIFAR-10 corners overlaid. The six Tier 1 corners (factorized, contrast, and **contrast_sdt**) track closely after ~ 50 to 80 epochs and continue improving slowly through the cosine schedule, finishing near 95.4 to 96.1%. The two pure **sdt** corners plateau earlier, with persistently higher train and validation loss, consistent with the diagonal restriction of SDSA (Section 3.6). Best epochs fall mostly in the 247 to 301 range under the 310-epoch schedule. Per-run panels are in Section A.1.

Table 5: CIFAR-10: published Spiking Transformer references vs. our best corners ($T=4$). Architecture notation L - D = depth-embed dim. Our Acc. is best official-test accuracy; published Acc. are as reported in the cited papers (typically test). Caveat: epochs/augs are not identical.

Method	Architecture	T	Acc. (%)
Spikformer Zhou et al. [2022]	4-384	4	95.19
Spikformer Zhou et al. [2022] [‡]	4-384 (400 E)	4	95.51
SDT Yao et al. [2023]	2-512	4	95.60
STAtten + Spikformer Wang et al. [2024]	4-384	4	94.36
STAtten + SDT Wang et al. [2024]	2-512	4	96.03
STAtten + QKFormer Wang et al. [2024]	4-384	4	95.35
A ² OS ² A Li et al. [2025]	2-384	4	95.70
A ² OS ² A Li et al. [2025]	2-512	4	96.42
Ours (STAtten-like)	2-512, factorized, bin.	4	96.10
Ours (Contrast-bin.)	2-512, contrast, bin.	4	95.65
Ours (HP-STAtten)	2-512, factorized, hyb.	4	95.57

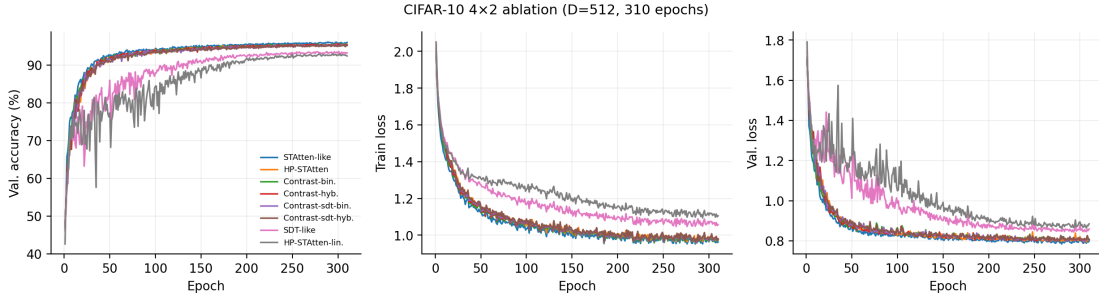


Figure 6: CIFAR-10 learning curves for all eight corners of the 4×2 ablation (Optuna t27 HPs transferred, $D = 512$, 310 epochs, seed 42; curves from MLflow). **Left:** official-test accuracy (%). **Center:** train loss. **Right:** validation loss. Factorized, contrast, and `contrast_sdt` form a high-accuracy tier (~ 95.4 to 96.1%); pure Hadamard (`sdt`) corners remain ~ 2.5 to 3 points lower with higher loss. Per-run panels are in Section A.1.

4.4 CIFAR-10-DVS: event-based ablation

To probe event-driven dynamics, we evaluate the same attention structures on CIFAR-10-DVS under the STAtten/SDT-aligned recipe: $D=256$, resize 64×64 , $c=4$, learning rate 0.01, weight decay 10^{-4} , batch size 16, TET $\lambda=0.001$, seed 42 (Section 4.1.1). The binary STAtten-like reference uses 200 epochs; the hybrid structure sweep uses a 310-epoch schedule (same optimizer settings). Recall that temporal chunking applies only to **factorized**; other modes mix per timestep.

Binary reference. The STAtten-like corner (**factorized**, **hybrid_qkv=false**) reaches **80.26%** best hold-out accuracy at epoch 198, on par with SDT ($\sim 80.0\%$) and within ~ 0.8 points of STAtten+SDT ($\sim 81.1\%$).

Hybrid structure sweep. Table 6 reports the four hybrid corners (**hybrid_qkv=true**). Unlike CIFAR-10, the ranking is *not* factorized-dominated: **contrast** leads at **81.55%** (above STAtten+SDT, 2-256), followed by hybrid **sdt** at 81.15%, then hybrid **factorized** at 79.96% and hybrid **contrast_sdt** at 79.66%.[†] Thus on event streams, cheap Hadamard attention can outperform factorized channel mixing under the shared DVS recipe, and VCA-light factorized contrast is the strongest hybrid corner. Figure 7 shows the corresponding learning curves.

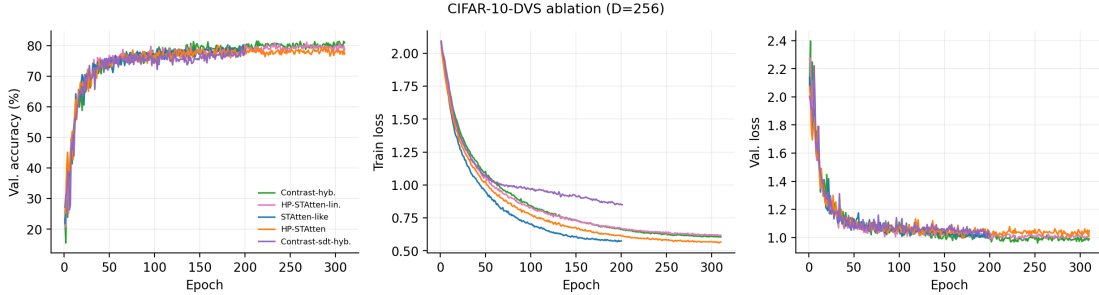


Figure 7: CIFAR-10-DVS learning curves from MLflow ($D=256$). Hybrid corners: 310 epochs; binary factorized reference: 200 epochs. **Left:** hold-out accuracy (%). **Center:** train loss. **Right:** validation loss. Contrast and **sdt** lead; hybrid factorized lags, unlike the CIFAR-10 ranking. **contrast_sdt** stops near epoch 202 (interrupted). Per-run panels are in Section A.2.

Precision axis on DVS (partial). For **factorized**, binary (80.26%) still edges hybrid (79.96%) despite the shorter binary schedule, consistent with the CIFAR-10 $\text{binary} \geq \text{hybrid}$ pattern for that structure. A complete binary column for **sdt/contrast/contrast_sdt** is not yet in the archived grid; we therefore do not claim a full DVS precision ranking.

Comparison to published neuromorphic results. Table 7 compares against the same reference papers (Spikformer, SDT, STAtten; A²OS²A does not report CIFAR-10-DVS in

Table 6: CIFAR-10-DVS ablation (2-256, $T=16$, 64×64 , 90/10 split). Hybrid corners: 310-epoch schedule; binary factorized reference: 200 epochs (not a matched-length ranking against the hybrid sweep). [†]`contrast_sdt` hybrid interrupted near epoch 202; best checkpoint already at ep. 196.

Variant	Mode	Hybrid	Schedule	Best hold-out (%)
Contrast-hyb.	contrast	✓	310 ep	81.55
HP-STAtten-lin.	sdt	✓	310 ep	81.15
STAtten+SDT (ref.) Wang et al. [2024]				~81.1
STAtten-like	factorized	✗	200 ep	80.26
SDT (ref.) Yao et al. [2023]				~80.0
HP-STAtten	factorized	✓	310 ep	79.96
Contrast-sdt-hyb. [†]	contrast_sdt	✓	~202 ep	79.66

the archived PDF). All listed models use $T=16$ and a 2-256 backbone (depth 2, embed dim 256) unless noted. Our contrast-hybrid corner (81.55%) exceeds STAtten+SDT (81.1%) and Spikformer (80.9%); hybrid `sdt` (81.15%) matches the STAtten+SDT band while remaining cheaper asymptotically than factorized mixing.

Table 7: CIFAR-10-DVS: published references vs. our corners ($T=16$, 64×64 , 90/10 hold-out). Numbers from the cited papers.

Method	Architecture	T	Acc. (%)
Spikformer Zhou et al. [2022]	2-256	16	80.9
SDT Yao et al. [2023]	2-256	16	80.0
STAtten + SDT Wang et al. [2024]	2-256	16	81.1
STAtten + SpikingReformer Wang et al. [2024]	4-384	16	80.60
STAtten + QKFormer Wang et al. [2024]	4-384	16	83.90
Ours (Contrast-hyb.)	2-256, contrast, hyb.	16	81.55
Ours (HP-STAtten-lin.)	2-256, sdt, hyb.	16	81.15
Ours (STAtten-like)	2-256, factorized, bin.	16	80.26
Ours (HP-STAtten)	2-256, factorized, hyb.	16	79.96
Ours (Contrast-sdt-hyb.)[†]	2-256, contrast_sdt, hyb.	16	79.66

[†]Interrupted near epoch 202; best at ep. 196. QKFormer+STAtten uses a larger 4-384 backbone than our 2-256 runs.

4.5 Complexity interpretation

Table 3 and Section 3.9 give the analytic ranking of attention-mixing cost. On CIFAR-10 after SPS ($N \approx 64$, $D_h = 64$, $n = 16$), the eight corners sit in FLOP families that only partially align with the accuracy tiers of Table 4:

- **Linear family** (`sdt`, `contrast_sdt`): $\sim 4 \times 10^3$ to 5×10^3 MACs/head $\rightarrow$ `sdt` at Tier 2 (~ 93.0 to 93.6%), but `contrast_sdt` at Tier 1 ($\sim 95.4\%$);
- **Factorized family** (`factorized`, `contrast`): $\sim 2.6 \times 10^5$ to 3.3×10^5 MACs/head $\rightarrow$ Tier 1 accuracy (~ 95.6 to 96.1%).

`contrast` costs roughly $1.25 \times$ `factorized` (dual $\pm$ streams) yet matches its accuracy under transferred HPs, supporting a *difference-over-similarity* bias at comparable asymptotic order. `contrast_sdt` stays cheap like `sdt` but, at $D = 512/310$ epochs, reaches Tier 1: the contrastive inductive bias appears to unlock much of the missing expressivity of pure diagonal aggregation under a stronger recipe. `hybrid_qkv` leaves this FLOP order unchanged; its intended benefit is an energy proxy (more AC-friendly encodings when binary), which we do not measure on hardware here. As N grows, Softmax MHSA ($\mathcal{O}(N^2 D_h)$) diverges from all grid modes, which remain linear in N . On event-based inputs, wall-clock training is often dominated by SPS rather than attention mixing; the complexity table is therefore primarily a *structural* and scalability argument, not a claim of GPU speedup.

4.6 Discussion

Six findings emerge.

1. **Training recipe and width first.** Moving from short unaugmented CIFAR-10 runs ($\sim 80\%$) to a $D=512$, 310-epoch Optuna-informed recipe (~ 95 to 96%) closes the gap to published Spiking Transformer numbers.
2. **Structure $\gg$ precision; binary $\geq$ hybrid.** Factorized, contrast, and `contrast_sdt` are nearly tied. Across all four structures, binary Q/K/V is at least as accurate as hybrid ($+0.01$ to $+0.60$ points). This *contradicts* the A²OS²A claim that hybrid encodings improve expressivity, but in a different setting: their gains were reported for naive $\mathcal{O}(N^2)$ token attention, whereas we use Softmax-free right-to-left maps ($\mathbf{K}^\top \mathbf{V}$) inside an SDT-style backbone, with HPs transferred from a hybrid-tuned Optuna run (which, if anything, should favor hybrid). In this regime, float keys / ternary values do not buy accuracy; binary remains preferable as both the accuracy and the neuromorphic energy proxy.
3. **Hadamard is cheaper but weaker here.** Pure `sdt` remains ~ 2.5 to 3.1 points below the Tier 1 peak; the gap may shrink after per-corner Optuna (Option A).
4. **Contrast factorized and `contrast_sdt` are both competitive at scale.** VCA-light with $\mathbf{K}^\top \mathbf{V}$ and its linear twin both reach Tier 1 under $D=512/310$ epochs, whereas pure SDSA does not, so differential tokens help when paired with either aggregation backend once capacity/schedule suffice. Because we already operate in the right-to-left $\mathbf{K}^\top \mathbf{V}$ regime, VCA’s usual complexity win *vs.* Softmax MHSA does not apply here: `contrast` sits in the same $\mathcal{O}((N+n)D_h^2)$ family as `factorized`, and the value of the integration is the inductive bias (and Softmax-free redesign), not an asymptotic leap.
5. **FLOPs $\neq$ accuracy $\neq$ energy.** Cheapest mixing (`sdt`) underperforms Tier 1; `contrast_sdt` shows that low FLOPs need not imply low accuracy when the

contrastive bias is present. Here energy and accuracy align on the precision axis (binary wins both under our proxies), but still diverge on the structure axis (cheap **sdt** is less accurate).

6. **Event-based ranking differs from CIFAR-10.** On CIFAR-10-DVS (hybrid sweep), **contrast** reaches **81.55%** and hybrid **sdt** **81.15%**, both above hybrid **factorized** (79.96%) and competitive with / above STAtten+SDT ($\sim 81.1\%$). The CIFAR-10 “factorized family dominates” story does *not* transfer to event streams under this recipe. Binary factorized remains slightly ahead of its hybrid twin (80.26% *vs.* 79.96%); completing the binary DVS column is needed for a full precision claim.

We deliberately avoid claiming neuromorphic *energy* gains until AC/MAC measurements under spike sparsity are available.

5 Conclusion and Perspectives

We presented HP-STAtten, a Spiking Transformer attention module that combines STAtten’s chunk-wise factorized attention with A²OS²A’s hybrid Q/K/V encodings, embedded in a spike-driven convolutional backbone. A single implementation exposes a 4×2 ablation grid over attention structure (factorized / Hadamard / contrast / contrast_sdt) and encoding precision. All integrations stay Softmax-free and *right-to-left* associative ($\mathbf{K}^\top \mathbf{V}$ then $\mathbf{Q}$); VCA-light keeps the contrastive inductive bias of original VCA while redesigning its left-to-right Softmax stages for SNN compatibility. On CIFAR-10, with Optuna trial #27 hyperparameters transferred across all eight 4×2 corners ($D=512$, 310 epochs, seed 42), factorized, contrast, and **contrast_sdt** variants reach ~95.4 to 96.1% official-test accuracy, while pure Hadamard (**sdt**) variants lag by ~2.5 to 3.1 points (~93.0 to 93.6%). Binary Q/K/V matches or slightly exceeds hybrid in all four structure pairs: a negative transfer of A²OS²A’s hybrid-advantage claim into the STAtten-style right-to-left regime. On CIFAR-10-DVS ($D=256$; partial grid), a hybrid structure sweep reaches up to **81.55%** (contrast), with hybrid **sdt** at 81.15% outperforming hybrid factorized (79.96%); the binary factorized reference is 80.26%. Event-based rankings therefore differ from CIFAR-10: Hadamard and contrastive modes are competitive or best under the DVS recipe.

5.1 Limitations

- Single seed (42) and transferred HPs: the ranking is informative but not fully fair across architectures (linear corners may be HP-mismatched).
- CIFAR-10-DVS: hybrid structure sweep is available, but the binary column is complete only for **factorized**; **contrast_sdt** hybrid was interrupted near ep. 202.
- No measured FLOPs/energy (AC *vs.* MAC under spike sparsity) per corner.

5.2 Future work

- Per-variant Optuna (Option A), especially for **sdt** / **contrast_sdt**.
- Complete remaining CIFAR-10-DVS binary corners and multi-seed confirmation of top corners.
- Hierarchical designs (linear early layers, factorized/contrast late layers).
- Complete orthogonal complexity ablations on the HP-STAtten reference (low-rank mixing, local windows) and measure AC/MAC energy proxies.
- Condensed CVPR submission drawing on the same experimental core.

5.3 Summary of internship deliverables

The internship produced: (i) modular reimplementations of four Spiking Transformer baselines; (ii) the HP-STAtten attention module with **factorized**, **sdt**, **contrast**, and **contrast_sdt** paths plus hybrid/binary Q/K/V; (iii) a shared training and logging framework; (iv) CIFAR-10 and CIFAR-10-DVS ablations; and (v) a documented complexity

analysis for follow-up work. Per-run learning curves are collected in Section A. Together, these deliverables provide a platform for studying the trade-off between attention expressivity, temporal modeling, and computational cost in Spiking Transformers.

References

- Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- Hongmin Li, Hanchao Liu, Xiangyang Ji, Guoqi Li, and Luping Shi. CIFAR10-DVS: An event-stream dataset for object classification. *Frontiers in Neuroscience*, 11:309, 2017.
- X. Li et al. A²OS²A: Addition-only spiking self-attention. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- Yifan Pu, Jixuan Ying, Qixiu Li, Tianzhu Ye, Dongchen Han, Xiaochen Wang, Ziyi Wang, Xinyu Shao, Gao Huang, and Xiu Li. Linear differential vision transformer: Learning visual contrasts via pairwise differentials. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Kaushik Roy, Akhilesh Jaiswal, and Priyadarshini Panda. Towards spike-based machine intelligence with neuromorphic computing. *Nature*, 575(7784):607–617, 2019.
- Yan Wang et al. STAtten: Spatial-temporal attention for spiking transformers. *arXiv preprint arXiv:2409.19764*, 2024.
- Man Yao, Liwei Hu, Zhaofei Zhou, Tiejun Li, Bo Xu, and Guoqi Fan. Spike-driven transformer. *arXiv preprint arXiv:2307.01694*, 2023.
- Zhaofei Zhou, Yue Zhu, Ye He, Yan Wang, Xiurong Ren, Rong Yan, Jian Zhang, and Tiejun Li. Spikformer: When spiking neural network meets transformer. *arXiv preprint arXiv:2209.15425*, 2022.

A Supplementary Material

This supplementary material provides per-run learning curves for the ablations in Section 4, exported from the HP-STAtten MLflow tracking database. The overlays in Figures 6 and 7 remain the primary visuals for comparison; the panels below support detailed inspection of each logged run.

A.1 CIFAR-10 (4×2 grid)

Protocol: Optuna trial #27 hyperparameters transferred to all eight corners, $D=512$, 310 epochs, seed 42. Each row shows train loss, validation loss, and official-test accuracy for one corner.

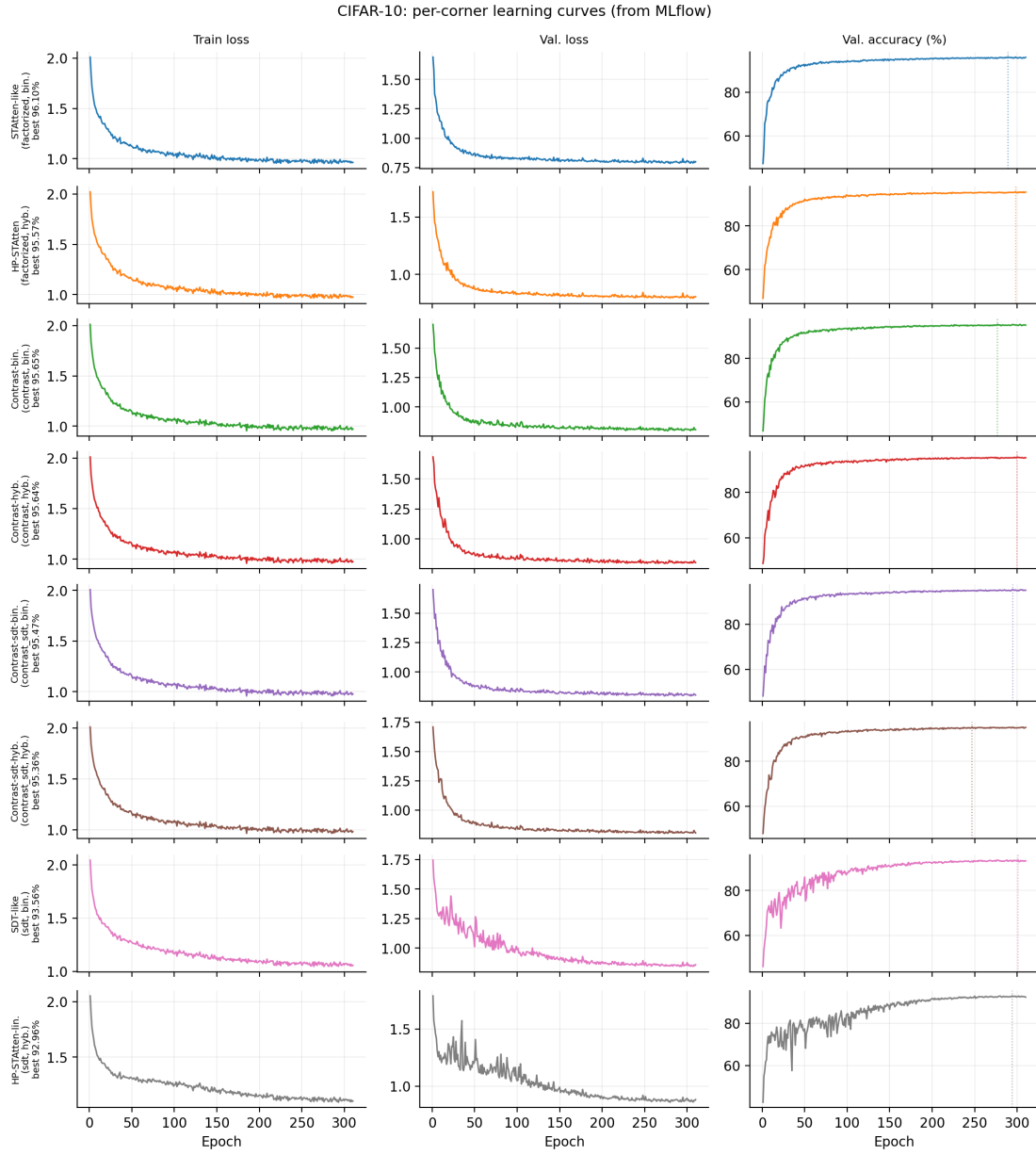


Figure 8: CIFAR-10 per-corner curves from MLflow ($D=512$, 310 epochs). Best official-test accuracy is annotated on the left of each row.

A.2 CIFAR-10-DVS

Protocol: $D=256$, $T=16$, 64×64 , 90/10 split. Hybrid corners use a 310-epoch schedule; the binary factorized reference uses 200 epochs. The `contrast_sdt` hybrid run was interrupted near epoch 202 (best checkpoint already at ep. 196).

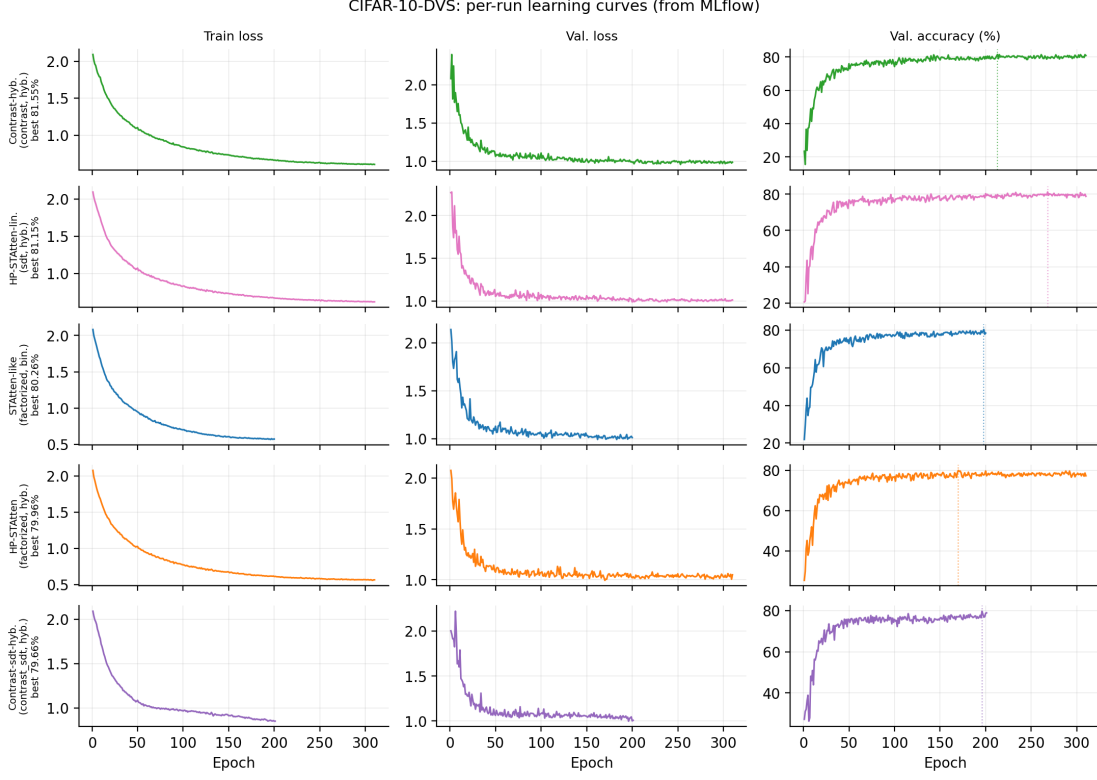


Figure 9: CIFAR-10-DVS per-run curves from MLflow. Best hold-out accuracy is annotated on the left of each row.